

# Pharmaceutical Manufacturing Case Study: Penicillin

How industrial fermenters, process energy, and equipment combine to produce an antibiotic at scale

|         |              |         |                                        |
|---------|--------------|---------|----------------------------------------|
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| Course  | ME 215, NJIT | Type    | Individual research project            |
| Product | Penicillin   | Reactor | Stainless steel fermenter (bioreactor) |

## PROJECT SUMMARY

This case study examines how penicillin is manufactured, with a focus on the reactor that makes it possible. Penicillin is produced by growing the mold *Penicillium chrysogenum* in large stainless steel fermenters that hold anywhere from 10 L in the lab to 100,000 L in industry. The study covers the raw materials, the sequence of processes from sterilization to packaging, the thermal and electrical energy each step requires, the major equipment, the by-products, and the design features — agitation, baffles, aeration, sensors, and temperature jackets — that let a fermenter hold the precise, sterile conditions the organism needs.

## 1. Product & Reactor

Penicillin is produced primarily in **fermenters**. These vessels are built from stainless steel, which can be repeatedly steam-sterilized and resists corrosion, allowing manufacturers to guarantee sterility and prevent contamination of the culture.

## 2. Raw Materials

Table 1. Key raw materials

| Material                       | Role in the process                                   |
|--------------------------------|-------------------------------------------------------|
| <i>Penicillium chrysogenum</i> | Production organism that synthesizes penicillin       |
| Glucose                        | Carbon and energy source for fungal growth            |
| Ammonium salts                 | Nitrogen source                                       |
| Phenylacetic acid              | Precursor that directs production toward penicillin G |
| Water                          | Growth medium and process fluid                       |

### 3. Manufacturing Process

Converting the raw materials into finished penicillin involves a sequence of essential stages:

Table 2. Process sequence from raw materials to finished product

| Stage            | Description                                                                                    |
|------------------|------------------------------------------------------------------------------------------------|
| 1. Sterilization | Steam sterilization of the fermenter, growth medium, and tools to establish aseptic conditions |
| 2. Fermentation  | Controlled growth of <i>P. chrysogenum</i> at about 26 °C with agitation and aeration          |
| 3. Separation    | Centrifuging to remove fungal biomass from the broth                                           |
| 4. Extraction    | Solvent extraction of penicillin from the filtered broth                                       |
| 5. Purification  | Chromatographic purification to reach pharmaceutical grade                                     |
| 6. Drying        | Removal of residual solvent and moisture                                                       |
| 7. Packaging     | Final dosage-form filling and packaging                                                        |

### 4. Energy Requirements

**Thermal energy.** Heat is essential for sterilizing the fermenter, the growth medium, and tools through steam sterilization, which establishes aseptic conditions. Thermal control then maintains the optimal temperature of about 26 °C for fungal growth throughout fermentation.

**Electrical energy.** Electricity powers the sensors, purification equipment, and the stirring and aeration systems. Motors drive the impellers that mix the broth, and pumps circulate solvent during extraction.

### 5. Equipment & By-products

The main production equipment includes fermenters, centrifuges, solvent extractors, chromatographic columns, and drying systems. The principal by-products are spent growth medium, used organic solvents, and fungal biomass.

### 6. Fermenter Design Characteristics

Table 3. Key design features of an industrial fermenter

| Characteristic          | Details                                                                   |
|-------------------------|---------------------------------------------------------------------------|
| Material                | Stainless steel, for sterility and excellent corrosion resistance         |
| Capacity                | About 10 L for laboratory units up to 100,000 L for industrial production |
| Agitators               | Impellers for thorough mixing of the broth                                |
| Baffles                 | Improve blending efficiency by preventing vortex formation                |
| Aeration ports          | Supply a steady flow of oxygen for microbial growth                       |
| Sensors                 | Continuously monitor pH, temperature, and dissolved oxygen                |
| Heating/cooling jackets | Provide precise temperature regulation during fermentation                |

## 7. Purpose of the Reactor

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Fermenters provide a precisely controlled environment that promotes optimal growth of *Penicillium chrysogenum*. By regulating temperature, pH, oxygen supply, and agitation, they allow manufacturers to maximize penicillin yield efficiently. Just as importantly, they maintain sterile conditions, minimizing the risk of contamination and protecting the integrity of the entire manufacturing process.

## References

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